The I_{3322} quantum value is attained spatially but not in finite dimension

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Abstract

Let $S = \omega_{\text{tensor}}(I_{3322}) = \omega_{\text{commuting}}(I_{3322})$ denote the quantum supremum of the I_{3322} Bell functional in the Collins–Gisin normalization (classical bound 0; two-qubit maximum exactly 1/4). Working from an exactly certified window $S \in (0.2508753845015185, 0.250875388108398]$ — a certified input built on Mghirbi's proof-carrying enclosure — and a certified equality of the tensor-product and commuting-operator suprema, we prove: (i) **no finite-dimensional quantum strategy attains** S — any finite local dimensions, pure or mixed states, projective or POVM measurements — proving the conjecture of Pál and Vértesi (2010); (ii) S **is** attained by a spatial strategy on $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ — infinite-dimensional attainment, which those authors asserted for their own construction, here established on an independent route. Consequently $C_q(3,3;2,2)$ is not closed — the smallest scenario by input count, among two-outcome bipartite scenarios, in which nonclosure is known — and $C_{qs}(3,3;2,2) \setminus C_q(3,3;2,2) \neq \emptyset$, settling the I_{3322} attainment question explicitly raised by Dykema, Paulsen, and Prakash. The nonattainment proof runs through a concave critical Bellman storage, exact rational endpoint-exclusion certificates, a reflection-gluing inequality, and a convex-envelope theorem excluding contact plateaus; finiteness then forces the two equality transports of an exact maximizer to coincide, which caps its value at $1/4 < S$. The attainment proof disintegrates the scalar spectral measure of a maximizing commuting state over the orbits of the group generated by its two response transports, and obtains an ℓ^2 Jacobi eigenvector whose normalizability is inherited from the probability measure rather than imposed. An earlier claimed proof by this program was publicly decertified after an exact audit; nothing decertified re-enters the present chain. We then determine the resulting dimension complexity: writing S_d for the optimum over strategies of local dimension at most d and $D(\varepsilon) = \min\{d : S - S_d \leq \varepsilon\}$, we establish $D(\varepsilon) = \Theta(\log(1/\varepsilon))$ — reaching S to accuracy ε requires and suffices local dimension of order $\log(1/\varepsilon)$. The upper half is proved in the paper, constructively, with the explicit derived bound $D(\varepsilon) \leq 23.9010650 \log(1/\varepsilon)$ (log natural) for all sufficiently small ε; the matching lower half, with existential constants, is established in a certificate chain shipped with the paper. The combinatorial and scalar cores of both halves are machine-checked in Lean 4 — cores, not the reductions. We close with a structural observation suggested by the results: in every case known to us, finite-dimensional attainment is accompanied by closed (finitely recurrent) optimal carriers, while the known transport-class nonattainment phenomena are carried by infinite open chains.

1 Introduction

The I_{3322} inequality [1, 2] is the best-known bipartite Bell functional beyond the Clauser–Horne–Shimony–Holt family: three binary measurements per party, and a facet of the local polytope inequivalent to CHSH. In 2010, Pál and Vértesi [3] computed its quantum value numerically to high precision, exhibited a family of finite-dimensional strategies converging to $0.2508753845\ldots$, observed that the qubit maximum is exactly $1/4$ and that within their construction no local dimension below 12 exceeds it, and stated in their abstract that their family "enables us to attain the largest possible quantum value in an infinite dimensional Hilbert space" [3]. They then posed the following, verbatim:

> "Taking as a conjecture that our construction is optimal (in the sense that no finite dimension suffices to achieve the true quantum maximum for I_{3322}), to our knowledge this constitutes the first example of a [bipartite] Bell scenario concerning [a] finite number of measurements and outcomes, where measuring finite dimensional quantum systems is not enough to obtain exactly all quantum correlations. ... We pose it as a challenge to provide an analytical proof of our conjecture." [3]

The parenthetical is a definition, not a second conjunct: to conjecture that the construction is "optimal" is, in their own gloss, to conjecture that no finite dimension suffices. Theorem 1 below proves exactly that statement. Their separate *assertion* of infinite-dimensional attainment is established here too, by Theorem 2, though on a route independent of their construction; whether their particular family converges to the true value, and the exact identification of that value, remain open (Section 6). As late as 2026, work coauthored by Vértesi still treats the I_{3322} infinite-dimensional optimum as unresolved [30]. Coladangelo and Stark, in constructing the first explicit bipartite correlation attainable only in infinite dimension (on larger alphabets), wrote of the I_{3322} case that "an analytical proof has remained elusive" [4] (we quote the Nature Communications version).

Theorem 1 (Nonattainment). *Let $S = \omega_{\text{tensor}}(I_{3322}) = \omega_{\text{commuting}}(I_{3322})$ be the quantum supremum in the normalization of Section 2, certified to lie in* $(0.2508753845015185, 0.250875388108398]$. *Then no finite-dimensional quantum strategy attains S: for all finite-dimensional Hilbert spaces $\mathcal{H}_A, \mathcal{H}_B$, every state ρ on $\mathcal{H}_A \otimes \mathcal{H}_B$, and every triple of binary measurements per party — projective or POVM, acting as effects of the form $F \otimes I$ on Alice's side and $I \otimes G$ on Bob's — one has $\mathrm{Tr}(\rho\,\mathcal{B}) < S$.*

Theorem 2 (Spatial attainment). *S is attained by a spatial strategy: there are projections $a_1, a_2, a_3 \in \mathcal{L}(\ell^2(\mathbb{Z}))$ and $b_1, b_2, b_3 \in \mathcal{L}(\ell^2(\mathbb{Z}))$, and a unit vector $\psi_S \in \ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$, such that setting $A_i = a_i \otimes I$ and $B_j = I \otimes b_j$ in (1) gives $\langle \psi_S, \mathcal{B}\,\psi_S\rangle = S$. In particular the resulting behavior lies in $C_{qs}(3,3;2,2)$.*

Here C_q denotes the set of bipartite correlations realizable with finite-dimensional tensor-product strategies, C_{qs} its analogue allowing arbitrary separable Hilbert spaces, and $C_{qa} = \overline{C_q}$; the taxonomy is that of Paulsen, Severini, Stahlke, Todorov and Winter [31], and we follow the presentation of [21, 4].

Corollary 3. *$C_q(3,3;2,2)$ is not closed, and $C_{qs}(3,3;2,2) \setminus C_q(3,3;2,2) \neq \emptyset$.*

The implication from I_{3322} nonattainment to nonclosure of $C_q(3,3;2,2)$ is immediate from the general compactness remark of Dykema, Paulsen, and Prakash [9], who explicitly raise the I_{3322} attainment question; the corollary settles it.

Among *two-outcome bipartite* scenarios, $(3,3;2,2)$ is the smallest venue by input count in which nonclosure is known. Within that class the previously smallest was Beigi's $(4,4;2,2)$ separation of C_q from C_{qs}, which implies nonclosure of C_q there [11]; the first two-outcome nonclosure witness was the earlier $(5,5;2,2)$ synchronous construction of Dykema–Paulsen–Prakash [8] (with a simplified proof by Musat–Rørdam [10]). Outside the two-outcome class the known witnesses are larger still: Beigi's $(4,4;3,3)$ nonclosure of C_{qs} [11], the $(4,5;3,3)$ correlation of Coladangelo and Stark [4], the embezzlement-based non-closure proof of [24], and the linear-system game of [13], whose input sets have sizes 184 and 235 and output sets sizes 8 and 2.

Three inputs per party is moreover the smallest possible venue among two-outcome scenarios, because a scenario in which *either* party has only two binary measurements has closed C_q. Indeed, Jordan's lemma simultaneously block-diagonalizes that party's two binary observables into blocks of dimension at most 2, so the behavior is a convex combination of the behaviors of the blocks; each block state then has Schmidt rank at most 2, so the other party's POVMs may be compressed to a two-dimensional support without changing $p(ab|xy)$. Hence $C_q(2,n;2,2)$ is the convex hull of a compact set, and closed, for every n; the case $(2,2;2,2)$ recovers the stronger statement that every functional there attains its maximum on two qubits [14]. Coladangelo and Stark had already singled the present scenario out: "the $(3,3,2,2)$ scenario is the simplest one that is suspected to separate C_q and C_{qs}" [4]. We do not claim minimality over scenarios with larger output alphabets.

Nonattainment opens a quantitative question that the qualitative theorems do not answer: how fast must local dimension grow as a strategy approaches S? Section 5 settles it. Writing S_d for the optimum at local dimension at most d and $D(\varepsilon) = \min\{d : S - S_d \leq \varepsilon\}$, Theorem 6 proves $D(\varepsilon) = \Theta(\log(1/\varepsilon))$: the upper half is constructive, truncating the attaining carrier of Theorem 2 to a window of length $O(\log(1/\varepsilon))$, and the lower half shows that nothing converges faster than geometrically. The cost of approaching the unattainable value is itself a theorem.

Correction history, stated first

This program previously released a claimed proof of the I_{3322} headline (exact optimum, spatial attainment, nonattainment) whose load-bearing global amplitude datum was subsequently *refuted by our own post-release exact audit*: the amplitude-compatibility equation of the original bi-infinite wall construction fails by an exactly certified margin (the mismatch lies in $[1.40 \times 10^{-4}, 1.79 \times 10^{-4}]$, excluding zero, reproduced by two independent interval engines). The claims were publicly decertified within a day, and the repository's release history preserves the full correction record. The results announced here were then rebuilt on independent routes; the decertified fixed point, amplitude profile, and tail constants appear on explicit *not-used* lists in the certificate dependency records, and the failed compatibility equation is not repaired anywhere in the present chain — the new attainment route never poses it (Section 4). We state this history first because computer-assisted claims are only as credible as their failure handling; the correction record is part of the result.

Relation to independent concurrent work

The nonattainment, nonclosure, and spatial-attainment results of this paper were first announced in public repository releases on 5 August 2026 (git tags v3.0.0 and v3.1.0 in [37]); the dimension-complexity rate followed on 7 August 2026 (v3.3.0). Frozen archival snapshots from that week are preserved under version DOIs 10.5281/zenodo.21826916 (v3.2.3, 6 August) and 10.5281/zenodo.21843326 (v3.3.0, 7 August); the concept DOI 10.5281/zenodo.21782008 resolves to the present merged paper of record (v4.0.0). Independently and concurrently, Pauwels [5] proved finite-dimensional nonattainment and nonclosure by a different route (spectral-weight matrices and a Jacobi-recurrence argument), and the concurrent work [6] independently establishes that the I_{3322} Bell inequality requires infinite dimensions. To our knowledge the three works were carried out independently — Pauwels states in print that nothing in his paper depends on this repository — and we regard them as concurrent. Beyond the shared core, the present paper additionally establishes spatial attainment on $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ and the dimension-complexity rate $D(\varepsilon) = \Theta(\log(1/\varepsilon))$ — the attainment question (posed by Pál–Vértesi and by Dykema–Paulsen–Prakash) and the convergence-rate question, both of which Pauwels' Section VI also records as open.

How the proofs work

Both theorems run through one reduction, which we state informally here and use throughout. It is convenient to replace each party's first two binary measurements by the pair

$$X = A_1 + A_2 - I, \quad Y = A_2 - A_1, \qquad U = B_1 + B_2 - I, \quad V = B_2 - B_1,$$

which for binary projections satisfy $X^2 + Y^2 = I$ and $XY + YX = 0$, and likewise for (U, V) — the pairwise Jordan structure of a pair of binary projections. In these coordinates (1) reads

$$\mathcal{B} \;=\; X \otimes U + \tfrac{1}{2} X \otimes I - \tfrac{1}{2} I \otimes U - I \;+\; Y \otimes (B_3 - \tfrac{1}{2} I) \;+\; (A_3 - \tfrac{1}{2} I) \otimes V,$$

so the third measurements enter only through the two off-diagonal blocks. Each spectral value $t \in [-1, 1]$ of X, or of U, is a *label*. A finite word of labels $\mathbf{c} = (c_0, \ldots, c_n)$ names the real symmetric tridiagonal (Jacobi) matrix $J_{\mathbf{c}}$ whose entries are explicit functions of neighbouring labels — diagonal $\delta(x, u) = xu + (x - u)/2 - 1$, off-diagonal $b(t) = \sqrt{1 - t^2}/2$ — and the Pál–Vértesi block-to-Jacobi identity realizes the Rayleigh quotients of $J_{\mathbf{c}}$ as Bell values of finite strategies. Then

$$S \;=\; \sup_{\mathbf{c}} \; \lambda_{\max}(J_{\mathbf{c}}),$$

the same supremum for the tensor and for the commuting model: that is input (T0) below, and it is why the two values agree for this functional.

Dualizing that supremum by an LDL/Schur-pivot computation on $qI - J_{\mathbf{c}}$ replaces the word supremum by a *one-site* problem. Call a positive continuous function g on the label interval a *storage*, and call it *Bellman-feasible at level* q when

$$\frac{b(x)^2}{g(x)} + g(u) \;\le\; q - \delta(x, u) \qquad \text{for all labels } x, u.$$

Feasible storages price the steps of a path: $b(x)^2/g(x)$ is charged to the edge leaving the label x and $g(u)$ to the edge entering u. Input (T0) says that the infimum of the levels admitting a feasible storage is exactly S, and it produces one such storage g_q at each level $q > S$, namely the infimum of terminal Schur pivots over finite histories ending at the given label. The object both proofs analyse is the *critical storage* g obtained from these as $q \downarrow S$, together with the *zero set* Z on which the scalar remainder of the resulting operator decomposition vanishes — the label pairs a strategy of value exactly S is allowed to occupy.

Theorem 1, in one paragraph. A strategy of value exactly S must occupy Z, and Z is the graph of a single strictly increasing injection P: each label is served by exactly one predecessor and serves exactly one successor. The occupied labels of a maximizer therefore carry two order-reversing maps — reflection $\sigma(u) = -u$, and $a = P^{-1} \circ (-P)$. A finite totally ordered set admits exactly one order-reversing bijection, so in finite dimension $a = \sigma$: every occupied label is paired with its own reflection. Eliminating amplitude ratios on such a reflected pair caps the value at $1/4$, and $S > 1/4$. Finite dimension is used exactly twice, and only the second use is irreducibly finite-dimensional (Remark 4).

Theorem 2, in one paragraph. The commuting model does have a maximizer, and the zero-set geometry above never used finite dimension, so the same two order-reversing maps act — now on the spectrum of the commuting pair of label operators X, U, which need not be finite. Their composition $\tau = a \circ \sigma$ is increasing and, off a null set, fixed-point free, so each of its orbits is order-isomorphic to $\mathbb{Z}$. Disintegrating the maximizer's scalar spectral measure over the orbits of the group generated by a and σ, and fixing one orbit, yields weights whose square roots are the amplitudes of an $\ell^2(\mathbb{Z})$ eigenvector of the Jacobi matrix at eigenvalue S. Normalizability is not imposed: it is inherited, because the weights came from a probability measure. Installing that eigenvector in the alternating block form returns a spatial strategy of value S.

Theorem 6, in one paragraph. Upper half: the labels along the carrier of Theorem 2 are monotone, so the amplitudes decay geometrically at each end, at rates κ_- and κ_+; truncating to a window costs only a two-bond boundary flux, and balancing the two ends gives local dimension of order $(1/\kappa_{\text{eff}}) \log(1/\varepsilon)$ with $\kappa_{\text{eff}} = (1/\kappa_- + 1/\kappa_+)^{-1}$. Lower half: an ε-good d-dimensional strategy has a small response defect, while a staircase edge budget caps its response walk at $O(d)$ steps; a defect too small for the available walk length would produce the reflected configuration that the $1/4$ ceiling forbids, so the defect — hence ε — cannot be smaller than $e^{-O(d)}$.

2 Setting and certified inputs

For binary effects $A_1, A_2, A_3, B_1, B_2, B_3$ (that is, $0 \le A_i, B_j \le I$; projections in the projective case), the Bell operator, in the normalization used throughout this paper, is

$$\mathcal{B} \;=\; -A_2 - B_1 - 2B_2 \;+\; A_1B_1 + A_1B_2 - A_1B_3 \;+\; A_2B_1 + A_2B_2 + A_2B_3 \;-\; A_3B_1 + A_3B_2, \quad (1)$$

which coincides, with no scaling or additive shift, with Eq. (3) of Pál–Vértesi [3] (equivalently, it reproduces their optimality conditions (9)–(14)). Transposing $A_1 \leftrightarrow A_2$ and $B_1 \leftrightarrow B_2$ carries (1) entry-for-entry onto the standard Collins–Gisin table — Alice marginals $(-1, 0, 0)$, Bob marginals $(-2, -1, 0)$, joint block $\left[\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 0 \end{smallmatrix}\right]$ — whose local (classical deterministic) bound is 0 [2]. The two-qubit maximum is exactly $1/4$ [3]; Pál–Vértesi further report, as a numerical finding of their dimension scan, that no construction they found below local dimension 12 exceeds $1/4$, with their strategy family converging to $0.250875384514\ldots$ [3]. We write ω_{tensor} for the supremum of $\mathrm{Tr}(\rho\,\mathcal{B})$ over strategies on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $\mathcal{H}_A, \mathcal{H}_B$ finite-dimensional, and $\omega_{\text{commuting}}$ for the supremum over states on the universal C^*-algebra generated by the two commuting triples. A *spatial strategy* is a tuple $(\mathcal{H}_A \otimes \mathcal{H}_B, \psi, \{a_i\}, \{b_j\})$ with $\mathcal{H}_A, \mathcal{H}_B$ separable, ψ a unit vector, and $\{a_i\}, \{b_j\}$ binary projective measurements on the respective factors acting as $a_i \otimes I$ and $I \otimes b_j$; the behaviors so realized are exactly C_{qs}. Throughout, S denotes the common value $\omega_{\text{tensor}} = \omega_{\text{commuting}}$ supplied by the certified input (T0) below, identified *only* by its certified window; we do not claim S equals the historical decimal, nor that the Pál–Vértesi family converges to S, though all certified data are consistent with both.

Exact rational two-sided enclosures for the I_{3322} quantum value predate our own releases: Mghirbi [7] published proof-carrying exact certificates with enclosure width below 10^{-9} — tighter than the window used here — in July 2026. Those certificates establish a strict enclosure; they do not address equality of the two models, attainment, or nonattainment, which are the subject of this paper. The window below is therefore a certified *input*, not a contribution.

The proofs consume the following exactly certified inputs, established in the public repository [37]: (T0) a Bellman/path variational theorem giving $\omega_{\text{tensor}} = \omega_{\text{commuting}} = S$, together with, for every level $q > S$, a terminal-pivot storage g_q — the infimum of terminal Schur pivots over finite histories — which is positive, Bellman-feasible, and equicontinuous in q; (LB) an exact 255-dimensional strategy certifying $S > 0.2508753845015185 > 1/4$ (rational square-root floors, independently reconstructed in 160-digit interval arithmetic); (UB) an exact rational 25,601-knot Bellman witness certifying $S \le 0.250875388108398$; and (W) a generic operator weld: any positive Bellman-feasible storage at level q yields a positive decomposition $qI - \mathcal{B} = R_0 + R_A + R_B$. The Pál–Vértesi block-to-Jacobi identity, in its no-endpoint alternating form, supplies the bridge between Jacobi data and Bell strategies; it is a two-site local identity, established symbolically and on exact rational fixtures, and independently replicated.

These are inputs, and the dependency runs one way. In particular (T0) is certified independently of the results below, and is not re-derived from them: Theorem 2 produces a spatial strategy of value $\omega_{\text{commuting}}$, which together with $C_{qs} \subseteq \overline{C_q}$ would give $\omega_{\text{commuting}} \le \omega_{\text{tensor}}$ — a consequence of (T0), not a premise of it.

3 Nonattainment

This section gives Theorem 1 at the level of its lemma chain: the five steps below are stated in the order they are used, each with the property of the critical storage it consumes. The complete arguments, with every constant exact, are in the proof documents of [37]; Section 7 records which parts are machine-checked.

The chain is: (N1) the critical storage exists and is concave; (N2) the two endpoints are excluded by exact rational margins; (N3) a hypothetical maximizer carries an exact equality module at S; (N4) on that module the occupied support is a strictly increasing one-to-one graph; (N5) in finite dimension the two order-reversing transports on that graph must coincide, which caps the value at $1/4$. Steps (N1), (N2) and (N4) never use finite dimension; (N3) uses it once, to reduce the limit passage to a finite interior spectrum, and (N5) uses it again in the closure step. Only the second of those two uses is irreducibly finite-dimensional (Remark 4), which is why Theorem 2 is possible.

(N1) The critical storage. As $q \downarrow S$ the storages g_q admit, by equicontinuity and Arzelà–Ascoli, a subsequence converging uniformly to a limit g, which is continuous, nonnegative, *concave* (each terminal pivot is affine in the target label, and an infimum of affine functions is concave), positive on the open interval, and Bellman-feasible at S. These properties, together with the reflection-gluing inequality established below, are all that is consumed. Concavity is

elementary but decisive: it forces every corner of the contact intercept downward, which is what ultimately excludes contact plateaus.

(N2) Exact endpoint exclusion. Two-edge finite histories give exact rational margins: taking the interior detour radius $r = 1/10$, the endpoint predecessor lines exceed the storage, uniformly over the certified window, by at least

$$m_+ = \frac{23686917837403}{3008753881083980} > 0.00787, \qquad m_- = \frac{274562305945801}{4008753881083980} > 0.0684,$$

only the positivity $m_\pm > 0$ is used below. Consequently the remainder R_0 has a uniform positive gap $m = \min(m_+, m_-)$ on all endpoint spectral sectors, so any state of value at least $q - \varepsilon$ carries endpoint mass at most ε/m, an exact maximizer at S carries none, and $x = \pm 1$ is never a Bellman predecessor at S. These are closed-form rational certificates; no branch geometry or reflected wing is used.

(N3) The equality module of a hypothetical maximizer. Suppose a finite-dimensional strategy attains S. (Purification is unnecessary: $\mathrm{Tr}(\rho R) = 0$ with $R \succeq 0$ gives $R\rho = 0$ directly. POVMs reduce to projective measurements by the extreme-effect replacement of the certificate documents: holding the state and the other effects fixed, each binary effect may be replaced by an extreme maximizer of its linear objective, and extreme binary effects are projections; this preserves the dimension and the tensor structure and does not decrease the value, hence preserves it, the original strategy being assumed to attain S.) Applying the weld at levels $q_n \downarrow S$ and using $\sum_\nu \mathrm{Tr}(\rho R_{\nu,n}) = q_n - S \to 0$, the endpoint gaps kill boundary mass first; on the remaining interior finite spectrum the coefficients converge, and the limiting positive remainders annihilate ρ. Thus the maximizer carries an exact critical equality module at S — *without* any claim that the storage infimum is attained (the endpoint boundary layer is bypassed, not solved).

(N4) Zero-set geometry. On the equality module, the scalar remainder symbol vanishes. Reflected finite histories give the gluing inequality $g(x)\,g(-x) \geq b(x)^2$ with $b(x)^2 = (1-x^2)/4$; full equality of the two Bellman inequalities and the Cauchy–Schwarz step forces $g(x)g(-x) = b(x)^2$ at every occupied pair, and a Monge (supermodularity) argument makes the full zero set monotone. The decisive step is a convex-envelope theorem: writing $C(x) = S+1-\frac{x}{2}-b(x)^2/g(x)$ on the open interval $(-1,1)$, Bellman feasibility $g(u) \leq C(x) + (\frac{1}{2} - x)u$ evaluated at the *target* $u = 0$ gives $C(x) \geq g(0) > 0$ for every source x, so the constant $g(0)$ is a convex minorant and the greatest convex minorant $\check{C}$ exists. (Should g vanish at an endpoint, concavity supplies a linear chord bound there while $b(x)^2$ vanishes linearly, so b^2/g stays bounded and C cannot blow down at the boundary.) Concavity of g makes every corner of C point downward, while a kink of $\check{C}$ at a contact point would force an upward corner. Hence $\check{C}$ is differentiable, no source serves two targets, and (by an exact involution $(x,u) \mapsto (-u,-x)$ of the constraint system, which requires no symmetry of g itself) no target is served by two sources. The occupied support of any maximizer is therefore a strictly increasing one-to-one graph.

(N5) Finiteness forces closure; closure caps the value. The equality module carries two response transports — scalar multiples of unitaries between complete fibres — implementing the decreasing involutions $a(u) = P^{-1}(-P(u))$ and $\sigma(u) = -u$ on the occupied support, where P is the increasing graph map just obtained; the graph property makes both total. A finite totally ordered set admits exactly one decreasing bijection, so $a = \sigma$ on the support: every occupied pair has its reflected partner. The resulting amplitude-ratio elimination (multiplicity-uniform: the transports are scalar multiples of unitaries between complete fibres) yields, for an occupied pair (x,u),

$$S \;=\; xu - 1 + \sqrt{\big(b(x)+b(u)\big)^2 + \tfrac{(x-u)^2}{4}} \;\leq\; -t + \sqrt{t} \;\leq\; \tfrac{1}{4}, \qquad t = 1 - xu,$$

contradicting $S > 1/4$. This proves Theorem 1. The nonclosure half of Corollary 3 follows by compactness of the behavior space: finite-dimensional behaviors approach value S; a convergent subsequence has value exactly S; closedness of C_q would give a finite-dimensional realization attaining S.

What is machine-checked here. The Lean 4 kernel checks the exact endpoint margins of (N2), the finite-closure lemma of (N5) — the single point at which finite dimensionality enters Theorem 1 — and the amplitude-elimination chain and quarter ceiling that (N5) ends with. Exact-arithmetic scripts guard the algebraic identities of (N1) and (N4), and each prints explicitly the analytic steps it does not cover: envelope existence and maximality, the gluing inequality, the limiting passage, spectral support, and operator closure (Section 7).

Remark 4. Finite dimension enters twice: once to reduce the limit passage above to a finite interior spectrum, and once in the closure step "a finite ordered set has one decreasing bijection." Section 4 replaces the first by a disintegration argument; only the second is irreducibly finite-dimensional, which is exactly why Theorem 2 is possible.

4 Spatial attainment

The idea. The commuting model always attains its value, and the zero-set geometry of Section 3 transfers to a commuting maximizer unchanged, because the envelope argument consumed only the storage properties and the gluing inequality. Of the two places where Section 3 used finite dimension, the limit passage is replaced below by exact operator blocks, while the closure step — the one that capped the value at $1/4$ — is the irreducibly finite-dimensional one, and that is what leaves room for attainment. What takes the place of a finite spectrum is a disintegration: the maximizer's scalar spectral measure decomposes over the countable orbits of the group generated by its two response transports, and a single orbit already carries an $\ell^2(\mathbb{Z})$ eigenvector of the Jacobi matrix at eigenvalue S. The point to watch is normalizability, which here is not imposed but inherited — the amplitudes are square roots of the weights of a probability measure. The route of our withdrawn release instead constructed amplitudes along a bi-infinite wall and then demanded a global amplitude-compatibility equation, the equation our own audit refuted; here no amplitude is ever constructed and that equation is never posed, so the obstruction is *dissolved* rather than repaired.

The chain is: (S1) a maximizing commuting state exists and the zero-set geometry of Section 3 transfers to it unchanged; (S2) the response transports are realized by exact operator blocks, giving Radon–Nikodym transport laws for the maximizer's scalar spectral measure; (S3) the response translation is fixed-point free off a null set, its orbits are order-isomorphic to $\mathbb{Z}$, and a Borel transversal is constructed explicitly; (S4) disintegration over the orbits of the group generated by the two transports yields, on one orbit, amplitudes that are automatically square-summable and satisfy the Jacobi eigenvalue equation; (S5) the Pál–Vértesi block identity installs the resulting eigenvector as a spatial strategy of value exactly S.

(S1) A commuting maximizer, and the same zero set. By weak-$*$ compactness of the state space of the universal commuting C^*-algebra, a maximizing state exists for the commuting model: the commuting-operator value of a Bell functional is always a maximum [19, 20], so nonattainment is strictly a tensor-model phenomenon. Passing to the GNS representation of that state, write Ω for the associated cyclic unit vector, so that expectations become vector expectations. The limit passage of Section 3 then applies with one essential modification — and the zero-set geometry of Section 3 transfers unchanged, since the envelope argument consumed

only the storage properties and the gluing inequality, neither of which used finite dimension; the occupied support of the commuting maximizer is again a strictly increasing one-to-one graph P, on whose labels the decreasing involutions $a(u) = P^{-1}(-P(u))$ and $\sigma(u) = -u$ act as in Section 3.

(S2) Transports without a finite spectrum. The modification: no finite spectral support is available, and the transports are expressed through the exact operator blocks $W = Y(B_3 - \frac{1}{2}I)$ and $W_B = (A_3 - \frac{1}{2}I)V$, which satisfy $W^2 = b(X)^2$ and $WX = -XW$, with the corresponding U-side identities for W_B, as projector identities (no global spectral involution is assumed; it need not exist). Here X and U are the commuting source and target label operators, and Y, V the corresponding reflection blocks. These yield exact Radon–Nikodym transport laws for the scalar spectral measure μ of the commuting pair (X, U): writing E for the joint projection-valued measure of that pair, $\mu(\Delta) = \langle \Omega, E(\Delta)\Omega \rangle$ is a Borel probability measure, normalized because $\|\Omega\| = 1$.

(S3) Orbits of the response translation. The fixed-point set of the response translation $\tau = a \circ \sigma$ carries no mass — on it the two transport densities coincide, which fires the quarter-ceiling elimination against $S > 1/4$; the set need not be empty, and nullity is all that is used. Off it τ is fixed-point-free and, being a composition of two decreasing maps, strictly increasing, so each of its orbits is order-isomorphic to $\mathbb{Z}$ and a Borel transversal can be *constructed* — by selecting the first rational separating an orbit point from its image — with no appeal to smoothness machinery.

(S4) Disintegration, and inherited normalizability. Passing to the group $G = \langle a, \sigma \rangle$ generated by the two transports, which is infinite dihedral and whose orbits may carry nontrivial stabilizers, μ disintegrates over the countable G-orbits. The transport laws are Radon–Nikodym identities, valid off a single null set of transversal points; its complement has full measure and is therefore non-empty, and fixing one point t there, the conditional measure μ_t satisfies them at every atom of its orbit. Interleaving the weights of the τ-orbit $\{u_n\}$ of t with its reflection $\{-u_n\}$ then produces labels (c_j) and amplitudes $\lambda_j = \sqrt{\mu_t(\{c_j\})}$ with, automatically, $1 \le \sum_j \lambda_j^2 \le 2$: *normalizability is inherited from the probability measure, not imposed* — this is the precise point where the historical obstruction disappears. The transport laws *derive* the cocycle $\lambda_{j+1}/\lambda_j = g(c_j)/b(c_j)$, and the Bellman contact equality closes the Jacobi eigenvalue equation $J\lambda = S\lambda$ exactly, where J is the Jacobi matrix whose entries §5.2 records; henceforth λ is rescaled to unit norm, which changes neither the cocycle nor the eigenvalue equation.

(S5) Installing the eigenvector as a strategy. The Pál–Vértesi block identity, in its no-endpoint alternating form, installs the normalized eigenpair as a spatial strategy with value S; the identity is two-site local, so it applies to the bi-infinite chain term-by-term, every Bell term being a sum over disjoint two-site blocks with diagonal sums dominated by $\sum_j \lambda_j^2$ and neighbour sums absolutely convergent by Cauchy–Schwarz. No limit is taken and no operator topology is invoked. Every finite principal block of J is itself $\preceq S\,I$, so the value cannot exceed S, and the eigenvalue equation attains it.

Concretely, the bipartite vector is $\psi_S = \left(\sum_j \lambda_j^2\right)^{-1/2} \sum_j \lambda_j \, e_j \otimes e_j$ in the standard basis of $\ell^2(\mathbb{Z}) \otimes \ell^2(\mathbb{Z})$ — a unit vector, as Theorem 2 requires; the alternating block form installs a_1, a_2, a_3 on the left factor and b_1, b_2, b_3 on the right, each an orthogonal direct sum of rank-one projections over one of the two perfect matchings of $\mathbb{Z}$, so that $A_i = a_i \otimes I$ and $B_j = I \otimes b_j$ as Theorem 2 requires. The block *form* here is Pál–Vértesi's; the labels c_j and the amplitudes λ_j come from the disintegration. This proves Theorem 2 and, with Theorem 1, the separation in Corollary 3.

What is machine-checked here. None of it. The measure-theoretic and operator-algebraic chains of this section are not formalized in Lean 4, and the exact-arithmetic guard scripts name the limiting passage, spectral support and operator closure explicitly among the steps they do not cover. They are proved here and in the certificate's proof document, and were audited in the review rounds recorded there (Section 7).

The measure-theoretic steps of this argument — the conull invariant set, the Borel transversal, and the uniqueness of the disintegration (§§6–9 of the certificate's proof document) — were originally flagged in the certificate's status file as carrying residual proof risk. That risk was discharged on 7 August 2026 by a full expanded write-up (25 numbered lemmas, each with complete proof and explicit quantifier labels, plus an axiom inventory) reviewed in two blind adversarial rounds: a hostile proof-surface review together with an independent countermodel search (22 constructed attacks, no counterexample), followed by a re-review of the repaired document that verified every repair item by item (`AMENDMENT-2026-08-07-SECTIONS-6-9` in the certificate directory). The status file itself is hash-frozen and deliberately unamended; the amendment is the correction of record.

Remark 5 (Envelope versus orbit). The closure argument of Section 3 forces the $u \mapsto -u$ symmetry only on the *occupied orbit* of a hypothetical finite maximizer; nothing forces the critical envelope itself to be reflection-symmetric, and no step above obtains one wing of the envelope by reflecting the other. The historical construction's implicit reflection symmetry is, in retrospect, adjacent to the exact incompatibility that forced its withdrawal.

5 Dimension complexity

This section determines how fast the finite-dimensional optima approach S. Throughout, log denotes the natural logarithm; and, in this section only, $\mu(t)$ denotes the band multiplier (not the spectral measure of Section 4), $\Delta(t)$ its diagonal value (not a Borel set), and E the response defect of [34] (not the joint spectral measure); the symbol I keeps its meaning as the identity operator throughout, and windows of integers are written $\mathcal{I}$. Define, for each $d \geq 1$,

$$\begin{aligned} S_d &:= \sup \big\{ \langle\psi|\mathcal{B}|\psi\rangle \ : \ \dim \mathcal{H}_A \leq d,\ \dim \mathcal{H}_B \leq d,\ \text{three projection-valued binary} \\ &\qquad\qquad \text{measurements per party on } \mathcal{H}_A,\ \mathcal{H}_B,\ \psi \in \mathcal{H}_A \otimes \mathcal{H}_B \text{ a unit vector} \big\}, \\ D(\varepsilon) &:= \min \{ d \in \mathbb{N} \ : \ S - S_d \leq \varepsilon \}. \end{aligned}$$

The parameter set is compact — a finite product of projection varieties on $\mathbb{C}^d$ together with the unit sphere of $\mathbb{C}^d \otimes \mathbb{C}^d$ — and the value is continuous, so the supremum defining S_d is attained; Theorem 1 therefore gives $S_d < S$ for every d. S_d is nondecreasing, and $S_d \to S$ by the construction below, so $D(\varepsilon)$ is well-defined and finite for every $\varepsilon > 0$. (The cap is symmetric on the two parties; the construction below delivers equal local dimensions, so the same function results.) The class is projective measurements and pure states, at local-dimension scope with no dilation; this loses no generality for the value: the objective is affine in the state, so an extreme — pure — state attains it, and affine in each binary effect separately, so each may be replaced by an extreme effect, which is a projection — the same replacement as in Section 3, and a specifically two-outcome fact. Naimark dilation is never invoked, so the local dimensions are unchanged [33]; in particular the POVM-indexed rate is the same function.

A priori there are three alternatives: $D(\varepsilon) = \Theta(\log(1/\varepsilon))$ — geometric convergence; or $D(\varepsilon)/\log(1/\varepsilon) \to \infty$, of which polynomial dimension cost $D(\varepsilon) \sim \varepsilon^{-c}$ is one instance and the $2^{\Omega(\varepsilon^{-1/8})}$ of [24] a far more expensive one; or $D(\varepsilon) = o(\log(1/\varepsilon))$, which super-geometric convergence such as $S - S_d \sim e^{-d^2}$ would produce. The following theorem selects the middle course by excluding both extremes, and it takes both halves to do so: windowing the attaining carrier converges at least geometrically, and — by the lower bound — nothing converges faster.

Theorem 6 (Rate)**.** *There are constants $c_1, c_2 > 0$ with $c_1 \log(1/\varepsilon) \leq D(\varepsilon) \leq c_2 \log(1/\varepsilon)$ for all sufficiently small $\varepsilon > 0$. The upper half holds with the explicit derived constant: $D(\varepsilon) \leq 23.9010650 \log(1/\varepsilon)$ for all sufficiently small ε.*

Where the two halves are proved. The upper half is proved in §5.2. The lower half is established in the certificate chain [34], under the same verification standard as the certificate chains behind Theorems 1 and 2; §5.1 maps its route and §5.3 records its inheritances — in particular, both halves use the statement, established in [37], that the zero locus Z is contained in a compact subset of $(-1, 1)^2$, at the standing set out in §5.3. The two halves are independent in route and in audit history.

In gap form, given the monotonicity of S_d: there is $\kappa_{\text{eff}} > 0$ (defined in §5.2, with $1/\kappa_{\text{eff}} \leq 23.9010650$) such that for every $0 < \alpha < \kappa_{\text{eff}}$ there is d_0 with $S - S_d \leq e^{-\alpha d}$ for all $d \geq d_0$; and there are $c, p, C > 0$ with $S - S_d \geq c(1 + d)^{-p} e^{-Cd}$ for every d.

5.1 The lower bound, imported

The lower half says no finite-dimensional strategy — windowed or not — approaches S faster than geometrically. It is established in the certificate chain [34] (eleven numbered documents, of which Docs 01–05 are cited by number below, with three combinatorial cores machine-checked in Lean 4, Section 7) and is not re-derived in this section. What follows is a route map of that argument, at the granularity a reader needs in order to see what the upper half is being matched against; where this subsection's wording and the chain's wording differ, the chain is the authority for its own statement. The statement established there is

$$S - S_d \;\geq\; c(1 + d)^{-p}\, e^{-Cd} \qquad \text{for every } d, \tag{2}$$

with existential $c, p, C > 0$, which gives $D(\varepsilon) \geq c_1 \log(1/\varepsilon) - O(1)$ — and the $-O(1)$ is absorbed into Theorem 6's clean form by shrinking c_1.

From an ε-good strategy to a small response defect. A d-dimensional strategy within ε of S induces, through the same scalar reduction that drives Theorem 1, a finite family of scalar labels on spectral cells satisfying the same-marginal response identities up to a global defect E bounded polynomially in ε: after the far-inversion and endpoint payments ([34] Docs 01–02), $E \leq C_0\, \varepsilon^{1/8}$ ([34] Doc 05, its display (5.1)). (The label-localization modulus appearing later in the chain is, by contrast, a compactness modulus with no claimed rate — the polynomial rate is claimed for E only.)

A small defect localizes an approximate fixed structure. By finiteness, the label family under the monotone response recursion contains a repeated cell — a δ-pseudo-cycle — and the first machine-checked core (Lean: `min_point_pseudocycle`) extracts from any cyclic δ-pseudo-orbit of a monotone real map a δ-approximate fixed point at its minimum, with the same accuracy δ, independent of cycle length. The certificate chain's min-point squeeze ([34] Doc 03) refines this to a reflected zero pair $(z, -z)$ of the response structure, by compactness (no rate at this step).

The quarter obstruction repels the fixed structure. At the reflected pair the chain derives neutral gain — the two parties' normalized transport multiplier gains coincide ([34] Doc 04, its display (4.1)) — which, combined with the chain's identification of its common transport value with S (Doc 04, display (5.3)), forces the distorted-return value $F(1) = 1/4$ for S itself, contradicting the certified $S > 1/4$. (Here $F(q) = (1+q)^2/(16q)$ is the certificate chain's distorted-return ceiling, $S \le F(q)$, with neutral gain forcing $q = 1$; only the qualitative fact $S > 1/4$ is consumed, not the window's numerical margin.) By contraposition, an ε-good strategy's labels can only *approach* the reflected configuration at a price: Doc 04's dimension-free constant bounds the approach in terms of the defect, $m_C^2 \le C_C\, E$, with m_C the chain's approach modulus and C_C dimension-free ([34] Doc 04, its display (6.1)).

Dimension caps the escape. The second machine-checked core (Lean: `staircase_card_le`, via `staircase_sum_injOn`) is the staircase edge budget: an inversion-free bipartite relation on $\texttt{Fin}\, m \times \texttt{Fin}\, n$ has at most $m + n - 1$ edges. After far-inversion deletion each parity class of the retained cell support is inversion-free, so a d-dimensional strategy's usable response graph has at most $4d - 1$ edges and its response walks have length at most $2d$ ([34] Doc 01 §4; Doc 05, display (1.3)). Along such a walk the chain compounds a per-step gain, and the third core (Lean: `prod_le_two_pow_sum`: a product of positive branch degrees is at most two to their sum) turns the walk's branch degrees — which enter as $D_k^{-1/2}$, with degree sum below $8d$, whence the exponent $4d$ — into the Green-product floor $\Gamma_d = 2^{-4d}\bar{\sigma}^{2d}$, where σ is the chain's per-step response constant and $\bar{\sigma} := \min(1, \sigma)$ ([34] Doc 02, its display (6.4)). The endgame ([34] Doc 05 §§3–5) plays the two against each other: a defect small compared to Γ_d would let the walk reach the forbidden reflected configuration, so $E \ge c'(1+d)^{-p'} e^{-C'd}$; the eighth-power inversion of the bound $E \le C_0 \varepsilon^{1/8}$ above then yields (2), the inversion converting the primed constants into (2)'s. Note where rates are and are not claimed: the two compactness-only steps (the label-localization modulus, and the refinement to an exact reflected pair) feed the endgame only through qualitative facts — the existence of the exact reflected configuration that the quarter ceiling forbids — while the quantitative chain runs entirely through the defect E and the dimension-free bound $m_C^2 \le C_C\, E$. This half was established through its own audit campaign on the certificate chain, independent in route and in audit history from the upper half's, though the two share the endpoint-positivity statement and the certificate's graph structure (§5.3).

5.2 The upper bound, explicitly

The upper half is constructive, and is given here in compressed form: the construction, the constants, and the route are all stated, while the full accounting of the endpoint-diagonal payment and its constants is carried by the proof document [35]. It consumes Theorem 2 and the inputs named below; the reductions, transport typing, and assembly are not formalized (§5.3).

The carrier. The proof of Theorem 2 supplies a positive unit vector $\lambda \in \ell^2(\mathbb{Z})$ — the *carrier* — and a bi-infinite label sequence $(c_j) \subset (-1, 1)$ with:

(i) the certificate's interior zero locus Z (the zero set of its scalar weld functional) is the graph of a strictly increasing one-to-one Borel map P, and $P(c_{j+1}) = c_j$ for every j — one map, every link (Theorem 2 certificate, §§6 and 10);

(ii) the transport law $\lambda_{j+1}/\lambda_j = g(c_j)/b(c_j)$ (Theorem 2 certificate, §11), where $b(t) = \sqrt{1-t^2}/2$ and g is the critical storage of Section 3, continuous and positive on the open interval (a uniform positive bound on the compact label set follows below);

(iii) the eigen-equation $J\lambda = S\lambda$ for the Jacobi matrix $J_{jj} = \delta(c_{j-1}, c_j)$, $J_{j-1,j} = b(c_{j-1})$, where $\delta(x, u) := xu + (x-u)/2 - 1$ is the certificate's diagonal cost;

(iv) every adjacent label pair lies in Z, and Z is compactly interior, $Z \subset\subset (-1,1)^2$.

Item (i) is proved in the certificate's §§6–9 (conull invariant set, Borel transversal, uniqueness of disintegration) and §10; an expanded write-up of §§6–9 is included in the same certificate directory as `AMENDMENT-2026-08-07-SECTIONS-6-9`. The compact interiority (iv) is the endpoint-positivity statement of [37], used here exactly as established there and at the standing set out in §5.3. The positivity in (ii), by contrast, needs no such input on the closed interval: what the argument below uses is a positive lower bound on the compact label set only, and that follows from Section 3's own properties of g — continuous, positive on the open interval — once (iv) confines the labels to a compact subset of it. Let K be the compact interior union of coordinate projections of the closure of Z; all labels and label limits lie in K, and on K the functions b, g, $r := g/b$ are continuous with $b \geq b_0 > 0$, $g \geq m_g > 0$, and r, $1/r$ bounded by $R_{\max} < \infty$.

Monotone labels and geometric tails. Since P is strictly increasing and one-to-one with $P(c_{j+1}) = c_j$ for every j, the labels satisfy $c_{j+1} = P^{-1}(c_j)$ and $c_{j-1} = P(c_j)$: the orbit is functional in both directions, so it is strictly increasing, strictly decreasing, or constant, and constancy at a single adjacent pair propagates to the whole sequence. The constant case is excluded: a constant label c would make λ exactly geometric with ratio $r(c)$ — not in $\ell^2(\mathbb{Z})$ in one direction if $r(c) \neq 1$, and a non-normalizable constant vector if $r(c) = 1$. Hence the whole sequence is strictly monotone, confined to the compact K, so both tail limits $t_\pm \in K$ exist, and by the transport law and continuity of r the outward one-step ratios converge:

$$y_+ := \lim_{j\to+\infty} \lambda_{j+1}/\lambda_j = r(t_+), \qquad y_- := \lim_{j\to-\infty} \lambda_{j-1}/\lambda_j = 1/r(t_-).$$

Dividing row j of $J\lambda = S\lambda$ by λ_j and passing to either limit gives

$$y + 1/y \;=\; \mu(t) := \frac{S - \Delta(t)}{b(t)}, \qquad \Delta(t) := \delta(t,t) = t^2 - 1, \tag{3}$$

at $t = t_\pm$. The elementary band identity — for $s = \sqrt{1-t^2}$,

$$\Delta(t) + 2b(t) \;=\; t^2 - 1 + s \;=\; s(1-s) \;\leq\; 1/4 \tag{4}$$

(machine-checked: `band_identity`, `s_mul_one_sub_s_le_quarter`, `band_quarter_ceiling`) — with $b \leq 1/2$ (`amplitude_b_le_half`) and the certified lower window endpoint $S_{\text{LO}} := 0.2508753845015185 < S$ yields, for every interior t,

$$\begin{aligned} \mu(t) - 2 \;&=\; (S - \Delta - 2b)/b \;\geq\; (S - 1/4)/b \;\geq\; 2(S - 1/4) \\ &>\; 2(S_{\text{LO}} - 1/4) \;=\; 0.001750769003037, \\ \text{so}\;\; \mu(t) \;&\geq\; \mu_{\min} := 2.001750769003037. \end{aligned} \tag{5}$$

Since ℓ^2 excludes the growing root of (3) at each end, both outward ratios equal the decaying root $x_{\text{dec}}(\mu) = 2/(\mu + \sqrt{\mu^2 - 4})$, strictly decreasing in μ, so uniformly

$$y_\pm \;\leq\; x_{\text{dec}}(\mu_{\min}) \;\leq\; 0.9590241, \qquad y_\pm^2 \;\leq\; 0.91972725 \;<\; 1. \tag{6}$$

In particular $r(t_+) = y_+ < 1$ while $r(t_-) = 1/y_- > 1$: the transport ratio crosses 1 strictly between the two tail limits — a nontrivial corroboration of the two-ended open-chain picture, at no extra cost. Tail mass sums amplitude squares, so with $\kappa_\pm := -2\log y_\pm$ the tails obey $\lambda_j^2 \leq A\, e^{-\kappa_\pm |j|}$ for some finite A, and — carrying the down-rounded surrogate $K_0 := 0.08367827985 \leq$

$-2\log x_{\mathrm{dec}}(\mu_{\min})$ (computed against the exact $x_{\mathrm{dec}}(\mu_{\min})$, not the rounded display in (6), which is too weak to support it), which is safe both in $K_0/2$ (rounded down) and in the denominator of $2/K_0$ (a down-rounded denominator over-estimates the quotient, the safe direction for an upper display):

$$\begin{aligned} \kappa_\pm \;&\geq\; K_0 = 0.08367827985 \text{ per index}, \\ \kappa_{\mathrm{eff}} \;&:=\; (1/\kappa_- + 1/\kappa_+)^{-1} \;\geq\; K_0/2 \;\geq\; 0.0418391, \\ 1/\kappa_{\mathrm{eff}} \;&\leq\; 2/K_0 = 23.90106493088\ldots \;\leq\; 23.9010650. \end{aligned} \tag{7}$$

(All displayed decimals are bounds rounded in the safe direction and re-verified in exact rational arithmetic by the repository's guard scripts, which re-derive every displayed constant from its stated inputs and fail on any mismatch. Because $1/\kappa_{\mathrm{eff}} \leq 2/K_0 = 23.90106493088\ldots$ sits strictly below 23.9010650 with a genuine gap, the eventual bound $D(\varepsilon) \leq 23.9010650 \cdot \log(1/\varepsilon)$ follows — a bare $\limsup \leq 23.9010650$ alone would not give it.)

The truncation, at exact local dimension d. For a window $\mathcal{I} = [-L, R] \cap \mathbb{Z}$, keep every full 2×2 rank-one block of each of the carrier's two alternating matchings whose paired indices lie in $\mathcal{I}$; at a severed endpoint pair, replace the block by the one-dimensional projector on the retained index (either of the two admissible one-dimensional assignments serves; we fix this one, and the choice is part of the endpoint-diagonal payment accounted below). All six measurement operators remain exact projections — each is a direct sum of orthogonal rank-one blocks and one-dimensional projectors (verified symbolically at arbitrary block angles for window sizes $m = 3, \ldots, 8$ by an independently written script; the general case is the direct-sum structure itself) — and the local dimension is exactly $d = |\mathcal{I}|$, with no dilation and no padding. The truncated state is the renormalized window of λ.

Error accounting. Let $v_{\mathcal{I}}$ be the value of the principal Jacobi compression on $\mathcal{I}$ and $V_{\mathcal{I}}$ the Bell value of the truncated strategy just built. The exact two-boundary flux identity (two lines from $J\lambda = S\lambda$):

$$S - v_{\mathcal{I}} \;=\; B_{\mathcal{I}}/M_{\mathcal{I}}, \qquad B_{\mathcal{I}} := b(c_{-L-1})\lambda_{-L-1}\lambda_{-L} + b(c_R)\lambda_R\lambda_{R+1},$$

with $M_{\mathcal{I}} = \sum_{j \in \mathcal{I}} \lambda_j^2$ the window mass — only the two cut edges are unpaid. Relative to the compression, the completion changes exactly three kinds of term: the two cut-bond off-diagonals (already and only accounted by the flux identity — not charged twice), the two retained endpoint diagonals, and nothing else. Each of the eight joint Bell terms has coefficient of modulus 1, and a joint term's diagonal is a product of two projector diagonals, so its change has modulus at most 2; the marginal coefficients contribute their absolute sum 4; hence the safe count $C_{\mathrm{diag}} \leq 4 \cdot 1 + 8 \cdot 2 = 20$ and

$$0 \;\leq\; S - V_{\mathcal{I}} \;\leq\; (1/2 + 20 \cdot R_{\max}) \cdot B_{\mathcal{I}}/(1 - T_{\mathcal{I}}),$$

with $T_{\mathcal{I}} := 1 - M_{\mathcal{I}}$ the omitted tail mass. (The bracketed constant is the proof document's safe count; its full derivation, including the conversion of endpoint weights through b_0, is in [35]. Any such multiplicative constant affects only the $O(1)$ term of $D(\varepsilon)$ and cannot touch the displayed 23.9010650, which is carried entirely by the decay rates.) Both $B_{\mathcal{I}}$ and $T_{\mathcal{I}}$ decay like $e^{-\kappa_- L} + e^{-\kappa_+ R}$ by the tail estimates above, so balancing $L : R$ as $\kappa_+ : \kappa_-$ with $|\mathcal{I}| = d$ exactly:

$$S - S_d \;\leq\; S - V_{\mathcal{I}_d} \;\leq\; \exp(-\kappa_{\mathrm{eff}} \cdot d + o(d)), \tag{8}$$

where the $o(d)$ absorbs the multiplicative constants, the integer rounding of L and R, and the ratio-to-tail conversion; therefore, for every $\eta > 0$, $D(\varepsilon) \leq (\kappa_{\mathrm{eff}} - \eta)^{-1} \log(1/\varepsilon)$ eventually; letting $\eta \downarrow 0$ gives $\limsup D(\varepsilon)/\log(1/\varepsilon) \leq 1/\kappa_{\mathrm{eff}} \leq 23.9010650$. This proves Theorem 6. $\square$

Remark 7 (Concrete dimensions). The geometric regime is visible at concrete sizes: the repository's independently written verification script evaluates explicit strategies of the Pál–Vértesi family — a lower bound on S_d, not the truncation above — on the sampled ladder $d \in \{3, 4, 6, 8, 12, 16, 24, 33\}$, with values exceeding the two-qubit ceiling $1/4$ at $d = 24$ and $d = 33$ (high-precision evaluation with rigorous Rayleigh-quotient lower bounds asserted by the script); by monotonicity of S_d, $S_d > 1/4$ for all $d \geq 24$. The script's fixed parametrisation is not Pál–Vértesi's tuned family, so no conclusion about $d < 24$ follows from these evaluations, and nothing is claimed about the exact crossing point; in particular there is no tension with Pál–Vértesi's own dimension-scan report, which concerns their tuned family.

5.3 Scope, and what is not claimed

Constants. The constants in the Θ are existential. The single explicit number 23.9010650 bounds the upper half's constant only; it derives from the certified window and the band algebra alone via a certified exact-rational chain, and is certainly not sharp (the true carrier rate is governed by r at the actual tail limits; the repository's earlier dimension-gap document [33] derives, on the now-decertified wall route and against a constant of that decertified layer, the sharper family-specific exponent $\log R = 0.075192859195702\ldots$ along its carrier family — per unit of local dimension, the same units as κ_{eff}, so the comparison $\log R > \kappa_{\text{eff}}$-bound is meaningful — a historical comparison only, consistent with and not an input here). The lower half's constants c, p, C are existential and no explicit values are claimed, so no explicit two-sided window on $D(\varepsilon)/\log(1/\varepsilon)$ is asserted; the $o(d)$ in (8) is not made explicit, and the Theorem's inequalities are claimed for all sufficiently small ε.

Machine-checking boundary. The Lean kernel checks the combinatorial and scalar cores named above (27 theorems, standard axioms only). The reductions, transport typing, and assembly are *not* formalized; they are carried by the certificate documents.

Inheritance. Both halves use one statement established in [37]: the zero locus Z is contained in a compact subset of $(-1, 1)^2$ — equivalently, occupied labels stay away from ± 1. It is proved there in full, from the reflection-gluing inequality, the Lipschitz modulus of g, the certified window, and exact rational endpoint reserves; the certificate records it as a *proved candidate*, meaning that its argument is complete and assumes no input beyond those already in use, while its own audit left open the provenance and typing of those inputs. We use it exactly as established there, and neither re-derive nor upgrade it. Both halves also consume the certificate's §6/§10 graph structure, whose formerly disclosed residual risk was discharged by the §§6–9 amendment named above; the carrier vector λ itself is consumed by the upper half only. The two halves are independent in route and in audit history, not in these shared roots. The repository's earlier dimension-gap document [33] had established a constructive one-sided rate for its specific carrier family together with a conditional lower bound whose missing step it named explicitly; the certificate chain [34] discharges that step, subject to the shared inheritances just recorded.

Priority, stated against the nearest prior art. Two qualifications are owed, and we state them rather than leave them to be found. First, the Ω-type dimension bounds already in the literature — for instance the $2^{\Omega(\varepsilon^{-1/8})}$ of [24] — attach to witnesses of the *distinct* separation $C_{qs} \neq C_{qa}$, and [24] gives a lower bound only, with no matching upper bound or tightness claim. Second, and more directly: Coladangelo and Stark remark, in the arXiv version of [4] immediately following their non-attainability theorem, that repeated application of their Schmidt-orbit maps exhibits an infinite geometric sequence and that "this can be used to obtain some quantitative bounds on the dimension required to induce a correlation close to the ideal one"; they decline to prove it, on the stated ground that more useful bounds exist for the $C_{qs} \neq C_{qa}$ separation. That remark does not appear in the published version. We take it at face value: for their ideal state, whose Schmidt coefficients are exactly geometric, an $O(\log(1/\varepsilon))$ upper bound is a routine truncation argument, and their own discussion already names truncation to bounded-energy subspaces as the approximation mechanism. What we claim is therefore narrower and, we believe, accurate: a first two-sided characterization established with complete proofs — the upper half in §5.2, the lower half in the certificate chain [34] — for any witness of $C_q \subsetneq C_{qs}$. We note also what their method does not immediately give: the present carrier is *not* geometric but asymptotically geometric at two different rates at its two ends, which is why the constant is the harmonic-type combination $\kappa_{\text{eff}} = (1/\kappa_- + 1/\kappa_+)^{-1}$ rather than a single decay rate — that two-ended structure, forced by $r(t_+) < 1 < r(t_-)$, is the technical content of §5.2. The claim is made to our knowledge, on a negative literature search — evidence, not a theorem of priority.

6 Discussion: attainment and open carriers

The two theorems exhibit a sharp mechanism: *finiteness forces the equality transports to close (return), and closure caps the value strictly below S; the value above the closed ceiling is carried only by an infinite open chain.* It is natural to ask how general this picture is. In every case known to us, a Bell functional with proven finite-dimensional attainment of its global supremum closes for an identifiable algebraic reason: bipartite correlator (XOR) functionals — including the chained inequalities — via Tsirelson's Clifford construction [15, 16] (anticommutation is two-step closure); all $(2, 2; 2, 2)$ functionals, whose extreme points are realized by qubit strategies [14]; fixed CGLMP functionals, whose global optimum is attained at local dimension d (proven for $d = 3, 4$ [17]; we make no minimality claim); and, informally, those families whose sum-of-squares certificates force finite-dimensional algebraic relations on optimal strategies.

Conversely, both known explicit transport-class nonattainment objects are infinite open ladders: the Coladangelo–Stark state is an exactly geometric ladder whose nonattainment proof (Theorem 12 of the arXiv version of [4]) is an explicit no-return orbit argument on the Schmidt spectrum, and the Pál–Vértesi family is a half-infinite Jacobi chain with offset 2×2 blockings [3] — the same architecture, on smaller alphabets. Coladangelo and Stark suggested that the block-diagonal form of the conjectured optimal I_{3322} measurements, resembling that of their own construction, makes the inequality "potentially amenable to ideas and techniques from our work"; Theorem 1 may be read as an answer to that suggestion, reached by a transfer-operator route. Embezzlement-based nonattainment [22, 23, 24] is carried by the log-flat ladder $p_j \propto 1/j$ — an open chain of a different decay class; whether this heuristic correspondence with its operator-algebraic classification [25] can be made precise is an open question we do not pursue here.

Group-theoretic and type-based obstructions [12, 8, 10, 26] form a third family outside this transport picture; Slofstra's undecidability of perfect-strategy existence for linear-system games [12], the RE-completeness of approximating the quantum value [26], and the arithmetical-hierarchy separation [27] — deciding whether the quantum value of a two-player nonlocal game equals 1 is Π_2-complete (Mousavi–Nezhadi–Yuen), whereas deciding whether the commuting-operator value equals 1 is Π_1 (Slofstra, as restated there), precisely because the finite-dimensional supremum need not be attained — imply that any general structural criterion must be restricted to a subclass.

Within the two-outcome bipartite class, where pairwise Jordan structure generates the offset

2×2 chain architecture of the constructions above, one direction of a possible dichotomy is already established, and it is the mechanism of Theorem 1: if a Bell functional attains its supremum finite-dimensionally, then the response translation of an optimal equality module has a finite orbit — its critical transport is *finitely recurrent*. For I_{3322} this is what excludes attainment, since finite recurrence caps the value at $1/4 < S$. Whether finite recurrence conversely *suffices* for finite-dimensional attainment we do not know; nor do we have a definition of the critical transport for a general two-outcome functional, and supplying one is the first obstacle to testing it. We also note the marginal/correlator boundary: Tsirelson's theorem covers correlators only, and it is exactly the marginal terms through which I_{3322} escapes it.

Finally, two cautions from the literature, both concerning the *hierarchy* sense of "attain" rather than the strategy sense used in this paper: non-saturation of finite levels of the Navascués–Pironio–Acín hierarchy [18] is not evidence of strategy-nonattainment — doubly-tilted CHSH functionals near the critical line $\alpha + \beta = 2$, where the boundary curvature degenerates, are attained at $d = 2$ yet are not saturated by NPA level 10, as reported in [28] with the level-10 figure due to Gigena and Kaniewski [32] — and the NPA hierarchy need not attain the commuting value at any finite level [29] (whose headline result is the undecidability of deciding $\omega_{qc} > 1/2$, in their notation for the commuting-operator value).

Theorem 6 is the quantitative face of this picture: the value carried by the open chain is approachable, but only at geometric cost, $D(\varepsilon) = \Theta(\log(1/\varepsilon))$. For contrast, the embezzlement-based witness of [24] — a witness of the distinct separation $C_{qs} \neq C_{qa}$ — requires local dimension $2^{\Omega(\varepsilon^{-1/8})}$; I_{3322} therefore sits at the opposite, logarithmic end of the known spectrum for the cost of approaching a value that is never attained.

Two questions about I_{3322} itself remain open. The first is the exact identification of S beyond the certified window. The second concerns the Pál–Vértesi family: they asserted that it attains the quantum value in infinite dimension, and our audit of the historical route found that the natural amplitude-compatibility equation for that wall fails by a certified margin (the correction history in Section 1). Whether their family nonetheless converges to S — equivalently, whether $S = 0.250875384514\ldots$ — is not settled here; Theorem 2 reaches S on an independent route that never poses that equation.

7 Methods and data

Verifying this paper's claims

1. `pip install -r requirements.txt`, then `python certificate/release/verify_release.py` — frozen SHA-256 custody + exact rational window; ~4 seconds; exit 0.
2. Theorem 1: `certificate/production/theorem-N-four-receipts-at-S/` — signed statement, three rounds of refutation-first, model-based adversarial review (see the disclosure below), exact-arithmetic guards.
3. Theorem 2: `certificate/production/theorem-S-spatial-attainment-at-S/` — proof documents, review record, and the amendment of 2026-08-07 discharging the initially flagged §§6–9 residual risk (25 lemmas, two blind review rounds).
4. Theorem 6, the rate $D(\varepsilon) = \Theta(\log(1/\varepsilon))$: `certificate/production/rate-theta-log/` — both bundles with promotion-gate records.
5. Lean cores: `cd lean/I3322Kernel && lake build` — no `sorry`, standard axioms.
6. `VERIFY.md` at repository root: every claim mapped to its check, with 15-minute / 2-hour / full-audit reading paths.

Exact computation

All load-bearing computations are exact — rational arithmetic or outward-rounded interval arithmetic — and two-engine where numerical: independent reconstructions in separate code-bases, so that the 255-dimensional lower strategy, for instance, is verified in exact rationals and reproduced at 160 digits. They ship as machine-checkable certificates in the repository [37], with the complete correction record. The analytic steps are established in the proof documents; the guard scripts check algebraic identities and exact arithmetic only. A Lean 4 kernel [36] (`lean/I3322Kernel/`; 27 theorems, all reporting standard axioms only) machine-checks the quarter-ceiling chain of Section 3, the band algebra of §5.2, and the three lower-bound combinatorial cores of §5.1; the kernel is intentionally minimal — cores, not reductions.

Use of language models

Large language models were used throughout this work — for proof discovery, implementation, verification engineering, adversarial auditing, and editorial assistance — across multiple model families and separate, independently instantiated sessions, with the auditing sessions instructed to refute rather than confirm. The author conceived the research program and its hypotheses, set the verification standards and the preregistration and correction protocols, made all promotion, retraction, and publication decisions, and assumes sole responsibility for this work. The load-bearing content does not rest on model output: every load-bearing computation is exact — rational or outward-rounded interval arithmetic — and independently reimplemented, and the combinatorial and scalar cores are checked by the Lean kernel (above).

Data availability

Repository: `https://github.com/Apsiape/i3322-exact-wall`. The proofs referenced throughout are pinned to the archived release of record (v4.0.0), version DOI 10.5281/zenodo.22099128; the concept DOI 10.5281/zenodo.21782008 resolves to the latest release and is not the object cited here. The certificate directories, all under `certificate/production/`, are

```
theorem-N-four-receipts-at-S/
theorem-S-spatial-attainment-at-S/
rate-theta-log/
```

Competing interests

None.

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